Supplementary Material: Learning Battery Crack Segmentation from Complete, Partial, and Unlabeled FIB-SEM Data

Samia Mohinta*
Department of Physiology, Development and Neuroscience, University of Cambridge
Cambridge, United Kingdom
Neurobiology Division, MRC Laboratory of Molecular Biology
Cambridge, United Kingdom

Amoghavarsha Mahadevegowda*
Department of Materials Science and Metallurgy, University of Cambridge
Cambridge, United Kingdom
The Faraday Institution
Didcot, United Kingdom

The supplementary material documents the data preparation, supervision objectives, training procedures, and evaluation conventions used in the main paper. It also provides auxiliary class metrics, qualitative examples of complete-label scaling, and an analysis of frozen pretrained representations.

1 ADDITIONAL DATA AND PREPROCESSING DETAILS

The principal annotated dataset is a continuous FIB-SEM stack of a single extracted particle. Complete masks encode crack, particle, and outside specimen as three semantic classes. Partial major-crack masks contain manually marked crack pixels, unresolved specimen-interior pixels, and outside-specimen pixels. The inside-specimen region comprises the marked cracks and unresolved interior. The specimen boundary and positive crack marks therefore provide two forms of supervision from the same annotation.

The 256 complete annotations form 16 non-overlapping groups of 16 consecutive slices. Ten groups are assigned to training, three to validation, and three to testing, yielding 160, 48, and 48 slices. The partial annotations occupy the final 64 consecutive sections of the annotated stack, with the test group positioned between the partial-training and validation groups. Complete-label budgets contain one, two, four, eight, or ten training groups, corresponding to 16, 32, 64, 128, or 160 complete masks. These budgets are nested and use the same fixed group ordering throughout the experiments. The groups prevent the same section from appearing in multiple splits, although sections on opposite sides of a split boundary may remain correlated.

Labeled slices are padded at the bottom and right from 300×403 to 416×416 pixels. Padding is excluded from both training losses and evaluation, and predictions are cropped back to the native field before scoring. Unlabeled images retain their native dimensions and are padded only when required by a model. Grayscale inputs supplied to three-channel pretrained encoders are repeated across channels and normalized using the pretrained encoder's input statistics. The near- and far-domain unlabeled groups were assigned by reviewing the corresponding image stacks with the annotator. Each stack belongs to one group; the assignment does not rely on a numerical measure of domain distance.

2 ADDITIONAL ARCHITECTURE AND TRAINING DETAILS

The U-Net has four encoder downsampling stages followed by a bottleneck, with channel widths 32, 64, 128, 256, and 256. Decoder features are bilinearly upsampled and concatenated with the corresponding encoder features. Each convolutional block uses Group Normalization [24] with at most eight groups and a sigmoid-weighted linear unit activation [5]. The network contains approximately 3.3 million parameters.

For μsam [1], grayscale inputs are repeated across three channels, resized to 1024×1024, and normalized using the pretrained encoder's input statistics. The ViT-B encoder produces a $256 \times 64 \times 64$ feature map, which is decoded through four progressive upsampling stages. The frozen configuration updates only the semantic decoder, while the final-block configuration additionally updates the last transformer block. LoRA adaptation [7] inserts rank-eight updates with scaling factor 16 into the combined query–key–value attention projection while leaving the remaining encoder parameters frozen.

The compact U-Net, SwinUNETR, and μsam experiments use AdamW [13]. The learning rate is 10^{-3} for the compact U-Net, SwinUNETR, and the μsam decoder, and 10^{-4} for trainable μsam encoder parameters. Weight decay is 10^{-4}, batch size is 8, and the global gradient norm is clipped at 1.0. Training lasts for at most 200 epochs. In partial-supervision experiments, complete and partial batches alternate. Each epoch contains eight complete batches and, when applicable, eight partial batches.

SwinUNETR [6] uses a two-dimensional MONAI configuration with feature size 24 and approximately 6.3 million parameters. Its parameters are trained from random initialization. The architecture follows the same optimization, augmentation, and model-selection protocol as the compact U-Net, including the three random seeds.

The nnU-Net v2 [8] comparison uses its self-configured two-dimensional PlainConvUNet and training procedure. The configured network has seven encoder stages and uses 256×384 patches in the complete-only configuration and 256×320 patches in the partial-supervision configurations. The corresponding batch sizes are five and seven. Optimization uses stochastic gradient descent with initial learning rate 0.01, momentum 0.99, weight decay 3×10^{-5}, and a polynomial learning-rate schedule. Each model is trained for 1,000 epochs with deep supervision and nnU-Net's own augmentation and checkpoint selection.

*Corresponding authors: sm2667@cam.ac.uk and am2729@cam.ac.uk.

All nnU-Net comparisons use one prescribed training–validation partition rather than multiple random initializations. The complete-only configuration has 64 complete training and 48 complete validation sections. The partial-supervision configurations add 32 partial training and 16 partial validation sections. The complete-only and naively exhaustive configurations use the standard nnU-Net segmentation loss, while custom objectives for R1, R2-positive, and R3 retain the known specimen membership, positive crack pixels, or both. The nnU-Net objectives follow the same annotation distinctions as the compact U-Net objectives, without imposing identical loss weighting or optimization.

For the compact U-Net and SwinUNETR, geometric augmentation includes horizontal and vertical flips, right-angle rotations, rotations within ±15°, scaling between 0.85 and 1.15, translations up to 5%, and elastic deformation with probability 0.2. Geometric transformations are applied identically to images and their corresponding annotation tensors. Masks use nearest-neighbor interpolation, and pixels introduced outside the original image field are excluded from the loss. Intensity augmentation is applied only to images and includes brightness and contrast adjustment, gamma adjustment, Gaussian noise, Poisson-like noise, Gaussian blur, sharpening, local contrast variation, and slow intensity drift. nnU-Net uses its separate automatically configured augmentation procedure.

For compact U-Net experiments, including nnPU, and SwinUNETR experiments, seeds 42, 43, and 44 control model initialization, sample ordering, and augmentation. Data augmentation is further seeded using the run seed, image identity, and training epoch. Deterministic PyTorch operations are enabled where supported [14], and cuDNN benchmarking is disabled [3].

For the compact U-Net, SwinUNETR, and μsam, model selection uses the 48 exhaustively labeled validation slices. The monitored score is the average of mean crack Dice and hard clDice. The retained checkpoint is updated when the validation score improves by more than 0.001, and training stops after 15 consecutive epochs without such an improvement. The retained checkpoint is reloaded for test evaluation. Each μsam configuration uses seed 42; nnU-Net v2 follows its own model-selection procedure.

3 LOSS AND TEACHER–STUDENT IMPLEMENTATION

For complete masks, inverse-frequency cross-entropy weights are computed from the selected complete-mask training subset and normalized to have mean one. The Dice term assigns equal weight to crack, particle, and outside specimen. Let p_{ic} be the softmax probability of class c at pixel i, let y_{ic} be its one-hot target, and let $m_i \in \{0, 1\}$ indicate whether the pixel is valid. The class-specific soft Dice is

$$D_c = \frac{2\sum_i m_i p_{ic} y_{ic} + \epsilon}{\sum_i m_i p_{ic} + \sum_i m_i y_{ic} + \epsilon}, \qquad \epsilon = 10^{-6}. \tag{S1}$$

The complete-mask loss is

$$\mathcal{L}_{\text{full}} = 0.5\,\mathcal{L}_{\text{CE}} + 0.5\left(1 - \frac{1}{3}\sum_{c=0}^{2} D_c\right). \tag{S2}$$

Padded pixels and pixels introduced outside the original image field by augmentation have $m_i = 0$ and therefore do not contribute to either term.

3.1 Partial-label objectives

Let $p_c(x)$ denote the probability of class c at pixel x. The partial masks define specimen-interior pixels Ω_{inside}, outside-specimen pixels Ω_{outside}, and marked major cracks Ω_{major}. The specimen loss distinguishes specimen membership without declaring unmarked interior pixels negative for cracks:

$$\ell_{\text{spec}}(x) = \begin{cases} -\log p_{\text{outside}}(x), & x \in \Omega_{\text{outside}}, \\ -\log[p_{\text{crack}}(x) + p_{\text{particle}}(x)], & x \in \Omega_{\text{inside}}. \end{cases} \tag{S3}$$

The specimen loss $\mathcal{L}_{\text{spec}}$ averages this quantity across valid pixels in the partial mask. The major-crack loss supervises only the marked positives:

$$\mathcal{L}_{\text{major}} = -\frac{1}{|\Omega_{\text{major}}|}\sum_{x\in\Omega_{\text{major}}} \log p_{\text{crack}}(x). \tag{S4}$$

In the shared compact U-Net and SwinUNETR procedure, R1 uses $0.5\mathcal{L}_{\text{spec}}$, R2-positive uses $0.5\mathcal{L}_{\text{major}}$, and R3 combines both losses with the coefficients specified in Table 1 of the main paper. R2-naive instead treats all unmarked interior pixels as particle and applies the complete-mask loss.

For R3, coefficient screening used

$$\lambda_{\text{spec}} \in \{0.10, 0.25, 0.50\}, \qquad \lambda_{\text{major}} \in \{0.05, 0.10, 0.25, 0.50\}.$$

We screened the full grid with seed 42 and subsequently completed selected coefficient pairs for seeds 43 and 44. Among the pairs completed for all three seeds, $(\lambda_{\text{spec}}, \lambda_{\text{major}}) = (0.25, 0.10)$ had the highest mean validation score and is reported as R3-calibrated. The same completed runs are used both for this selection and for the reported three-seed summary, so the R3-calibrated result is not an independent post-selection estimate.

3.2 Positive–unlabeled loss

The non-negative positive–unlabeled (nnPU) comparison [9] uses the same compact U-Net and complete-mask supervision as R3. Within partial images, annotated major-crack pixels form the positive set P, while unmarked specimen-interior pixels form the unlabeled set U. With binary positive and negative logistic losses ℓ_+ and ℓ_- and assumed crack prior π, the risk is

$$\mathcal{L}_{\text{nnPU}} = \pi\,\mathbb{E}_P[\ell_+] + \max(0, \mathbb{E}_U[\ell_-] - \pi\,\mathbb{E}_P[\ell_-])\,. \tag{S5}$$

The partial-image objective is $0.5\mathcal{L}_{\text{spec}} + 0.5\mathcal{L}_{\text{nnPU}}$. The marked-crack loss is not added separately, since its positive information is already included in the nnPU risk. We evaluate $\pi = 0.02$, 0.05, and 0.10, as well as a prior estimated from the fraction of crack pixels among crack and particle pixels in the selected complete training masks ($\pi = 0.33587$). The nnPU objective uses unresolved pixels from partial masks, rather than additional unlabeled images.

3.3 Teacher–student consistency

R4 uses an exponential-moving-average teacher with decay 0.99 [21]. Following confidence-filtered weak-to-strong pseudo-labeling [19], the teacher receives a weak view with shared geometric augmentation and intensity clipping. The student receives a strongly intensity-augmented view with the same geometry. Far-domain student views additionally receive FIB-SEM artifact

augmentation. Let q_{ic} and p_{ic} denote teacher and student probabilities, respectively. The hard teacher label and confidence indicator are

$$\hat{c}_i = \arg\max_c q_{ic}, \qquad a_i = \mathbf{1}\left[\max_c q_{ic} \geq 0.90\right]. \tag{S6}$$

For near-domain images, retained pixels use their three-class teacher labels:

$$\mathcal{L}_{\text{near}} = -\frac{\sum_i a_i \log p_{i,\hat{c}_i}}{\max(1, \sum_i a_i)}. \tag{S7}$$

For far-domain images, teacher-outside predictions are excluded and consistency is restricted to crack versus particle. We define

$$z_i = \mathbf{1}[\hat{c}_i = \text{crack}], \qquad b_i = a_i\, \mathbf{1}[\hat{c}_i \neq \text{outside}], \tag{S8}$$

and use

$$\mathcal{L}_{\text{far}} = -\frac{\sum_i b_i \left[z_i \log p_{i,\text{crack}} + (1 - z_i) \log\left(1 - p_{i,\text{crack}}\right)\right]}{\max(1, \sum_i b_i)}. \tag{S9}$$

Thus, confident teacher-crack predictions provide positive crack targets, confident teacher-particle predictions provide negative crack targets, and teacher-outside predictions do not contribute to far-domain consistency.

R4 retains the R3-default partial-loss coefficients. Its total objective is

$$\mathcal{L}_{\text{R4}} = \mathcal{L}_{\text{supervised}} + w(t)\mathcal{L}_{\text{near}} + w(t)\mathcal{L}_{\text{far}}, \tag{S10}$$

where $\mathcal{L}_{\text{supervised}}$ is the complete-mask objective for a complete batch or the R3-default partial objective for a partial batch. The consistency coefficient $w(t)$ increases linearly from zero to one over the first 20% of the configured maximum training steps and remains at one thereafter.

To assess sensitivity to pseudo-label selection and update strength, we also evaluate confidence thresholds of 0.70, 0.80, and 0.95; final consistency weights of 0.25 and 0.50; an EMA decay of 0.999; and additional artifact augmentation. The reference configuration uses threshold 0.90, final weight 1.0, and EMA decay 0.99. All ablations use seeds 42, 43, and 44.

R4-default applies FIB-SEM artifact augmentation to far-domain student views. The artifact-augmentation ablation additionally applies the same transformations to complete and partial labeled training images, after standard geometric and intensity augmentation. The transformations comprise column-wise banding (probability 0.50, multiplicative strength 0.15), a vertical intensity gradient (0.40, range $[-0.10, 0.10]$), localized charging-like brightening (0.25, amplitude 0.25), and attenuation of one to three scan rows to 20% intensity (0.30). Augmentation of unlabeled views and the remaining R4 training settings are unchanged.

4 ADDITIONAL EVALUATION CONVENTIONS

Metrics are calculated separately for each test slice and then averaged within a run. Unless stated otherwise, U-Net summaries report mean and standard deviation across seeds 42, 43, and 44. For ensemble predictions, class probabilities are averaged across the three seeds before each pixel is assigned to its highest-probability class. For each three-seed comparison, 95% confidence intervals are computed from run-level means using $\bar{m} \pm t_{0.975,2}\, s/\sqrt{3}$, where s is the sample standard deviation and $t_{0.975,2} = 4.303$. Paired intervals use the three within-seed differences. These intervals describe variation across the evaluated random seeds, not independent particles. The nnU-Net configurations each have one completed training run and therefore have no seed-level confidence interval.

For a predicted crack mask P and exhaustive reference crack mask G, Dice [4] is

$$\text{Dice}(P, G) = \frac{2|P \cap G|}{|P| + |G|}. \tag{S11}$$

For clDice [17], let $\mathcal{S}(P)$ and $\mathcal{S}(G)$ denote one-pixel skeletons of prediction and reference. Topological precision and sensitivity are

$$T_{\text{prec}} = \frac{|\mathcal{S}(P) \cap G|}{|\mathcal{S}(P)|}, \qquad T_{\text{sens}} = \frac{|\mathcal{S}(G) \cap P|}{|\mathcal{S}(G)|}, \tag{S12}$$

and

$$\text{clDice} = \frac{2T_{\text{prec}} T_{\text{sens}}}{T_{\text{prec}} + T_{\text{sens}}}. \tag{S13}$$

If both predicted and reference crack masks are empty, Dice and clDice are defined as one. If exactly one is empty, both are defined as zero. All 48 exhaustive test slices used in the reported experiments contain reference crack pixels.

For predicted and reference crack masks P_i and G_i on test slice i, and the reference specimen region Ω_i, the crack fraction and absolute error are

$$\phi_i(M) = \frac{|\{x \in \Omega_i : M(x) = \text{crack}\}|}{|\Omega_i|}, \quad E_{\text{frac},i} = |\phi_i(P_i) - \phi_i(G_i)|. \tag{S14}$$

The crack-fraction error reported in the main paper is the mean across test slices. Prediction and reference use the same specimen region and denominator. For morphology, both crack masks are restricted to this same reference specimen region before skeletonization. Skeletons are computed with scikit-image [23], using Zhang–Suen thinning for these two-dimensional masks [25]. Skeleton length is the number of foreground skeleton pixels rather than a Euclidean arc-length estimate. Because an in-plane physical calibration is unavailable, length is reported in pixels.

Relative skeleton-length error for test slice i is

$$E^{\text{rel}}_{\text{skel},i} = \frac{|L^P_i - L^G_i|}{L^G_i}, \qquad L^G_i > 0, \tag{S15}$$

where L^P_i and L^G_i denote predicted and reference skeleton-pixel counts. We summarize this quantity using its median. For the morphology analysis in the main paper, predicted skeleton length is first averaged across the three U-Net seeds for each test slice. Spearman correlation [20], median relative error, and a Theil–Sen fit [15, 22] are then calculated between the resulting per-slice predictions and the reference lengths. Each test slice therefore contributes one observation.

For complete-label scaling, crack recall is calculated on each exhaustive test slice as $|P \cap G|/|G|$, where P and G are the predicted and reference crack regions. We average recall across the 48 exhaustive test slices.

Major-crack recall on the 16 held-out partial masks uses only manually marked major-crack pixels as reference positives. Unresolved specimen-interior pixels are ignored and define neither positive nor negative reference labels.

For the unlabeled-image audit in the main paper, normalized entropy [16] is calculated from the three-seed R0 ensemble trained

Table S1: Auxiliary particle and outside-specimen Dice at the fixed budget of 64 complete training masks. Values are mean ± standard deviation across seeds 42, 43, and 44. Bold indicates the highest value in each column.

Regime	Particle Dice	Outside Dice
R0	**.915±.003**	**.9953±.0004**
R1	.907±.008	.9940±.0006
R2-naive	.884±.012	.9905±.0030
R2-positive	.900±.008	.9930±.0013
R3-default	.898±.006	.9931±.0009
R3-calibrated	.908±.004	.9942±.0006
R4	.882±.013	.9902±.0019

with all 160 complete masks. If $\bar{p}_c(x)$ is the ensemble-averaged probability of class c at pixel x, normalized entropy is

$$H(x) = -\frac{1}{\log 3} \sum_{c=1}^{3} \bar{p}_c(x) \log \bar{p}_c(x). \tag{S16}$$

Binary disagreement is one when the hard prediction of this U-Net ensemble differs from that of the μsam LoRA model trained with 64 complete masks, and zero otherwise. Because these images do not have exhaustive reference masks, entropy and disagreement are diagnostic quantities rather than measures of accuracy.

5 AUXILIARY CLASS METRICS

The main paper focuses on crack-region and crack-network measures. For completeness, Table S1 reports Dice for the particle and outside-specimen classes in the heterogeneous-supervision experiment. These quantities are retained as auxiliary diagnostics and are not used to rank the crack-segmentation configurations.

6 QUALITATIVE COMPLETE-LABEL SCALING

The three-seed probability ensembles improve from 16 to 160 complete masks on all 48 test slices. The mean per-slice Dice gain is 0.052, with a range from 0.045 to 0.059, while the mean clDice gain is 0.035, with a range from 0.028 to 0.047. To avoid selecting examples by visual preference, Figure S1 shows slices at the 25th, 50th, and 75th percentiles of the per-slice Dice improvement.

7 FROZEN-FEATURE ANALYSIS

In addition to the segmentation models compared in the main paper, we examine how frozen image representations separate the three semantic classes through clustering and linear probes. Features are extracted from the μsam ViT-B encoder [1], DINOv3 ViT-S/16 [18], and the SAM3 vision encoder [2]. For μsam, we use the $256 \times 64 \times 64$ image-encoder output. For DINOv3, we concatenate patch representations from the final, fourth-from-final, and seventh-from-final transformer layers. For SAM3, we use the finest vision-encoder feature-pyramid output. Each feature map is bilinearly upsampled to the image canvas and cropped to the native image field.

All frozen-feature analyses use the same split as the main segmentation experiments, with 160 complete-mask training slices, 48 validation slices, and 48 test slices. Random seed 0 is used throughout. Validation data are not used to select PCA, clustering, or linear-probe hyperparameters.

7.1 PCA visualization

For each encoder, principal component analysis is fitted separately to a class-balanced sample from all 160 complete training slices. We sample up to 12 pixels per class from each slice, yielding at most 1,920 pixels per class and 5,760 pixels in total. If a slice contains fewer than 12 pixels of a class, all available pixels from that class are retained.

PCA is applied directly to the sampled feature vectors without prior standardization. Thus, the operation centers the features but does not rescale individual feature dimensions to unit variance. A three-component projection is fitted with random seed 0. For visualization, each component is scaled independently using its 1st and 99th percentile values and mapped to one RGB channel.

Figure S2 visualizes spatial variation in each frozen representation rather than semantic predictions. Because PCA is fitted and scaled separately for each encoder, colors are meaningful only within a panel. Similar colors across different encoders do not indicate corresponding feature directions or semantic classes.

7.2 K-means clustering

Three-cluster K-means [12] is fitted to the same class-balanced training sample used for PCA. We use the raw frozen feature vectors without standardization or unit-length normalization, with 10 initializations and random seed 0.

Because cluster identities have no intrinsic semantic meaning, we compute a one-to-one Hungarian assignment [10] using only the labeled training pixels to associate the three clusters with crack, particle, and outside specimen. This assignment is then fixed and applied unchanged to the held-out test slices. No test labels are used to determine the cluster-to-class mapping.

7.3 Linear probes

For the supervised frozen-feature analysis, separate multinomial logistic-regression probes are trained using either 64 or all 160 complete training masks. From each selected training slice, we retain up to 1,000 pixels per class. If fewer than 1,000 pixels of a class are available, all available pixels are retained. This sampling supplies the class balance, so no additional class weights are applied.

Each feature dimension is standardized using the mean and standard deviation estimated only from the sampled training pixels, and the same transformation is then applied unchanged to test features. The classifier uses an ℓ_2 penalty, regularization parameter $C = 1.0$, an L-BFGS solver [11], tolerance 10^{-4}, and at most 2,000 iterations. Random state 0 is used throughout. Encoder weights remain fixed during both clustering and probe training.

Table S2 reports the held-out crack Dice of the frozen-feature analyses. Clustering does not isolate the crack class reliably, while linear probes yield higher overlap. Among the tested probes, SAM3 performs best at both labeling budgets. The PCA visualization serves a different purpose: it shows variation within the representations rather than semantic predictions.

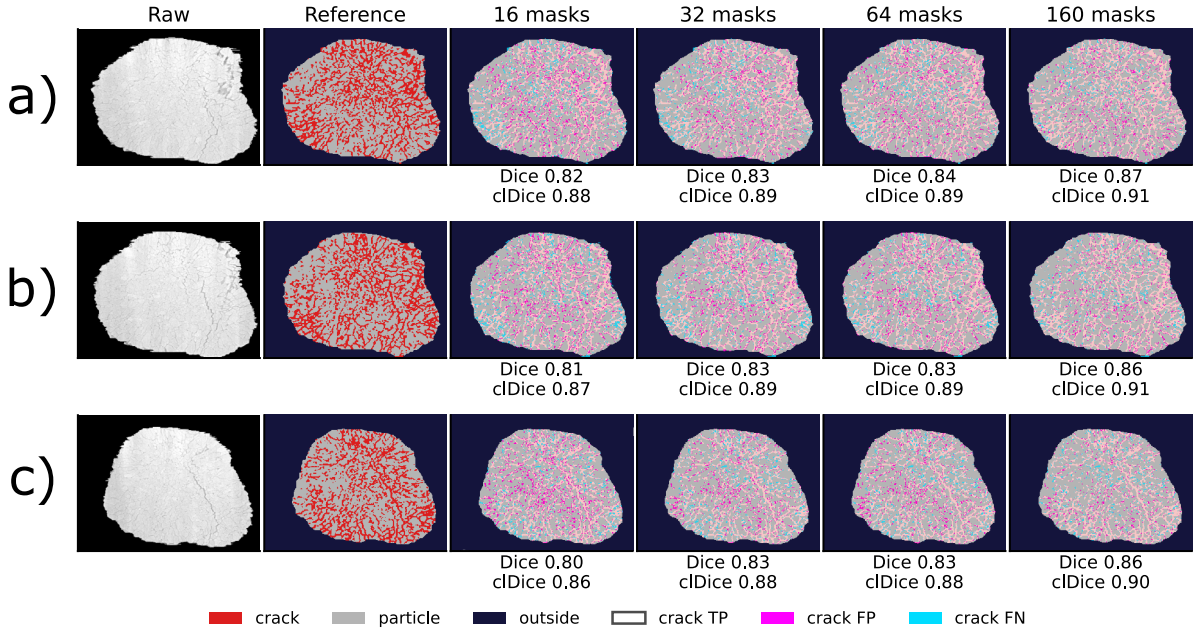


Figure S1: Qualitative effect of the complete-label budget. Rows show slices at the 25th, 50th, and 75th percentiles of the Dice improvement from 16 to 160 complete masks. White, magenta, and cyan denote crack true positives, false positives, and false negatives, respectively.

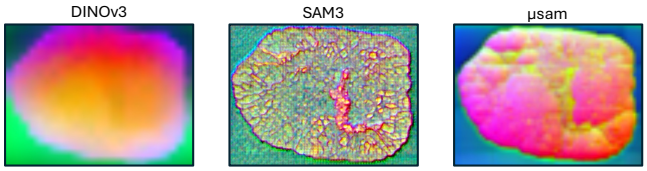


Figure S2: PCA visualization of frozen dense representations from DINOv3, SAM3, and μsam. Each encoder uses an independently fitted three-component PCA projection. Colors therefore show within-encoder feature variation and do not represent crack, particle, or outside-specimen classes.

Table S2: Held-out crack Dice for three-cluster K-means and frozen-feature linear probes. Probe budgets indicate the number of complete training masks.

Encoder	K-means	Probe, 64	Probe, 160
μsam	.386	.582	.581
DINOv3	.433	.467	.493
SAM3	.410	.678	.679

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